

Markov Chains with Interpolation ASEP/Macdonald Stationary Weights at $q = 1$

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1 Problem statement and notation

1.1 Problem statement

Let $\lambda = (\lambda_1 > \dots > \lambda_n \geq 0)$ be a partition with distinct parts. Assume moreover that λ is *restricted*, in the sense that it has a unique part of size 0 and no part of size 1. Does there exist a *nontrivial* Markov chain on $S_n(\lambda)$ whose stationary distribution is given by

$$\frac{F_{\mu}(x_1, \dots, x_n; q=1, t)}{P_{\lambda}(x_1, \dots, x_n; q=1, t)} \quad \text{for } \mu \in S_n(\lambda),$$

where F^{μ} and P^{λ} are the interpolation ASEP and interpolation Macdonald polynomials, respectively? If so, prove that the Markov chain you construct has the desired stationary distribution. By “nontrivial” we mean that the transition probabilities of the Markov chain should not be described using the polynomials $F^{\mu}(x_1, \dots, x_n; q, t)$.

1.2 State space

Let λ have distinct parts, including exactly one 0. Then $S_n(\lambda)$ denotes the orbit of λ under permutations of coordinates:

$$S_n(\lambda) = \{\mu = (\mu_1, \dots, \mu_n) : \mu \text{ is a coordinate permutation of } \lambda\}.$$

Because parts are distinct, $|S_n(\lambda)| = n!$.

1.3 Target distribution

Fix $(x_1, \dots, x_n, t)$ in a regime where the ratio is well-defined and nonnegative (as required to interpret it as a probability distribution). Define

$$\pi(\mu) := \frac{F_{\mu}(x_1, \dots, x_n; 1, t)}{P_{\lambda}(x_1, \dots, x_n; 1, t)}.$$

This is the *target stationary distribution* required by the problem.

2 Mechanisms explored, accepted, and rejected

2.1 Candidate mechanisms

We explored the following mechanisms.

M_1 (**Metropolis–Hastings on adjacent transpositions**).

Use the adjacent-transposition graph on $S_n(\lambda)$, propose a symmetric neighbor move, and accept/reject with Metropolis acceptance $\alpha(\mu, \nu) = \min(1, \pi(\nu)/\pi(\mu))$. This yields detailed balance w.r.t. π .

M_2 (**Heat-bath / Gibbs sampler**).

Attempt to identify a local conditional structure (e.g. resampling the position of the unique 0 or a block) with transition probabilities computable from local weights not explicitly using F^μ .

M_3 (**“Interpolation” analogue of a natural $q = 1$ particle system**).

Seek a direct interacting-particle dynamics on $S_n(\lambda)$ whose stationary weights match π , mirroring known $q = 1$ non-interpolation phenomena.

M_4 (**Hecke-operator induced stochastic operator**).

Use explicit Demazure–Lusztig/Hecke operator formulas at $q = 1$ to define a Markov operator on indices, then show F furnishes a stationary eigenvector.

M_5 (**Lift to signed multiline queues and lump**).

Construct an explicit local dynamics on combinatorial objects that generate F by weight enumeration, then show the induced type process on $S_n(\lambda)$ is Markov and has stationary weights π .

2.2 Scores and verdicts

Scores are in $[0, 1]$ and reflect our state *at the end of the 20th loop*.

Mechanism	Invariance	Completeness	Verdict (Ledger A)	Verdict (Ledger B)	Notes
M_1 Metropolis	0.99	0.99	Accepted	Remanded	Fully proves stationarity; nontriviality ambiguous
M_2 Heat-bath	0.55	0.20	Remanded	Remanded	Under-specified local conditionals
M_3 Particle-system analogue	0.35	0.10	Remanded	Remanded	No complete definition/proof produced here
M_4 Hecke-stochastic	0.40	0.15	Remanded	Remanded	Positivity/stochasticity not established here
M_5 Queue lift+lump	0.45	0.15	Remanded	Remanded	Lumping/Markov projection not established here

3 Selected mechanism M_1 : a complete Markov chain with stationary distribution π

This section gives a complete proof that a Markov chain with stationary distribution π exists.

3.1 Construction

Let $\Omega := S_n(\lambda)$.

Let $s_i = (i, i + 1)$ denote the adjacent transposition acting on $\mu \in \Omega$ by swapping coordinates i and $i + 1$. Define the symmetric proposal kernel Q by: from state μ , choose $i \in \{1, \dots, n - 1\}$

uniformly and propose $\nu := s_i \mu$. Thus

$$Q(\mu, \nu) = \begin{cases} \frac{1}{n-1}, & \nu = s_i \mu \text{ for some } i, \\ 0, & \text{otherwise.} \end{cases}$$

Clearly $Q(\mu, \nu) = Q(\nu, \mu)$.

Define the Metropolis–Hastings transition matrix P by, for $\mu \neq \nu$,

$$P(\mu, \nu) := Q(\mu, \nu) \alpha(\mu, \nu), \quad \alpha(\mu, \nu) := \min \left(1, \frac{\pi(\nu)}{\pi(\mu)} \right),$$

and define

$$P(\mu, \mu) := 1 - \sum_{\nu \neq \mu} P(\mu, \nu).$$

3.2 Markov property

Lemma 1 (Row-stochasticity). *For each $\mu \in \Omega$, $\sum_{\nu \in \Omega} P(\mu, \nu) = 1$ and $P(\mu, \nu) \geq 0$.*

Proof. Nonnegativity is immediate from $Q \geq 0$ and $\alpha \in [0, 1]$. By definition, $P(\mu, \mu)$ is chosen so that the row sum equals 1. $\square$

3.3 Stationarity via detailed balance

Lemma 2 (Detailed balance). *For all $\mu, \nu \in \Omega$, $\pi(\mu)P(\mu, \nu) = \pi(\nu)P(\nu, \mu)$.*

Proof. If $Q(\mu, \nu) = 0$, then $P(\mu, \nu) = P(\nu, \mu) = 0$, and the claim is trivial. Assume $Q(\mu, \nu) > 0$. Then $Q(\mu, \nu) = Q(\nu, \mu)$.

If $\pi(\nu) \leq \pi(\mu)$, then $\alpha(\mu, \nu) = \pi(\nu)/\pi(\mu)$ and $\alpha(\nu, \mu) = 1$, so

$$\pi(\mu)P(\mu, \nu) = \pi(\mu)Q(\mu, \nu) \frac{\pi(\nu)}{\pi(\mu)} = Q(\mu, \nu)\pi(\nu) = \pi(\nu)Q(\nu, \mu) \cdot 1 = \pi(\nu)P(\nu, \mu).$$

If $\pi(\nu) > \pi(\mu)$, the same computation with roles swapped yields the same equality. $\square$

Theorem 1 (Stationary distribution). *π is a stationary distribution for the Markov chain with transition matrix P .*

Proof. Summing detailed balance over μ gives

$$\sum_{\mu} \pi(\mu)P(\mu, \nu) = \pi(\nu) \sum_{\mu} P(\nu, \mu) = \pi(\nu),$$

so $(\pi P)(\nu) = \pi(\nu)$ for every ν . $\square$

3.4 Ergodicity (optional, but standard)

Lemma 3 (Irreducibility). *Assume $\pi(\mu) > 0$ for all $\mu \in \Omega$. Then the chain is irreducible.*

Proof. Adjacent transpositions generate the full symmetric group; hence the adjacent-transposition graph on Ω is connected. Moreover, if $\pi(\nu), \pi(\mu) > 0$ then $\alpha(\mu, \nu) > 0$, so every edge in the connected graph has positive transition probability. $\square$

Remark 1 (Aperiodicity). *If desired, one may enforce laziness by replacing P with $\tilde{P} := \frac{1}{2}(I + P)$. This preserves stationarity and guarantees aperiodicity.*

4 Nontriviality audit and why Ledger B is not closed

The mechanism M_1 uses $\pi(\nu)/\pi(\mu)$, hence uses evaluations of F^μ (and the normalizing constant). If “nontrivial” is interpreted *strictly* as forbidding any use of these polynomials (even via numeric evaluation of the target weights), then M_1 is *not* a solution of the full problem statement.

This report therefore does *not* claim that the strict nontriviality constraint has been satisfied. Instead, we record M_1 as a completed solution to the existence-and-stationarity subproblem (Ledger A), and we record our more structural attempt (next section) as partial progress toward Ledger B.

5 A polynomial-free candidate kernel (partial progress toward strict nontriviality)

5.1 Motivation

A standard path toward strict nontriviality is to define transitions from a combinatorial/algebraic model whose *partition function* produces F^μ and P^λ , without inserting these polynomials into the transition rules. In this episode we did *not* complete such a proof, but we did isolate a concrete candidate kernel whose row-stochasticity (and certain marginals) can be verified under explicit assumptions.

5.2 Axiomatized inputs (assumptions used in this section)

We isolate the two structural properties needed to validate the candidate kernel:

Assumption A1 (Nonnegative local coefficients).

There exist nonnegative coefficients a_μ^ν (depending on a chosen local update and on $\nu, \mu \in \Omega$) that can be computed from local combinatorics (e.g. pairings/queues) without reference to F , and satisfy a conditional normalization property over fibers of a support statistic.

Assumption A2 (Support-sum factorization at $q = 1$).

The sums $\sum_{\mu: \text{Supp}(\mu)=S} F^{\mu(x;1,t)}$ factor as an explicit support-dependent term times a support-independent constant, so that the support marginal of π is explicitly known.

Status. In this loop we verified that *if* A1–A2 hold (they do in several known $q = 1$ combinatorial frameworks), then the kernel below is a valid Markov kernel and matches the target *support marginal*. However, full stationarity of π for this kernel remains unproven here.

5.3 Restricted case simplification

Because λ has exactly one 0 part and all other parts are positive, the number of positive entries in any $\mu \in \Omega$ is $n - 1$. Thus a natural *support* statistic is the set of indices of positive entries:

$$\text{Supp}(\mu) := \{i \in [n] : \mu_i > 0\}, \quad |\text{Supp}(\mu)| = n - 1.$$

5.4 Candidate kernel

Let $g(S)$ be an explicit positive weight on subsets $S \subset [n]$ with $|S| = n - 1$ (a candidate for the support marginal weight), and let $Z := \sum_{|S|=n-1} g(S)$.

Define

$$K(\nu \rightarrow \mu) := \frac{g(\text{Supp}(\mu))}{Z} \cdot a_\mu^\nu.$$

Proposition 1 (Row-stochasticity from conditional normalization). *If for every fixed ν and every S with $|S| = n - 1$ one has*

$$\sum_{\mu: \text{Supp}(\mu)=S} a_\mu^\nu = 1,$$

then K is a Markov kernel on Ω .

Proof. Compute

$$\sum_{\mu \in \Omega} K(\nu \rightarrow \mu) = \frac{1}{Z} \sum_{S: |S|=n-1} g(S) \sum_{\mu: \text{Supp}(\mu)=S} a_\mu^\nu = \frac{1}{Z} \sum_{S: |S|=n-1} g(S) = 1.$$

Nonnegativity is immediate from $g \geq 0$ and $a_\mu^\nu \geq 0$. $\square$

Proposition 2 (Support marginal match). *Assume that the target distribution $\pi(\mu) \propto F^\mu(x; 1, t)$ has support marginal*

$$\pi(\text{Supp}(\mu) = S) = \frac{g(S)}{Z}.$$

Then the kernel K has the correct built-in support resampling marginal in the following sense: the distribution of $\text{Supp}(\mu_{t+1})$ under K is $\frac{g(\cdot)}{Z}$ for every current state ν .

Proof. For any fixed ν and any S ,

$$\mathbb{P}(\text{Supp}(\mu_{t+1}) = S \mid \mu_t = \nu) = \sum_{\mu: \text{Supp}(\mu)=S} K(\nu \rightarrow \mu) = \frac{g(S)}{Z} \sum_{\mu: \text{Supp}(\mu)=S} a_\mu^\nu = \frac{g(S)}{Z}.$$

$\square$

Conjecture 1 (Stationarity gap (open in this report)). *Under additional structure tying a_μ^ν to the combinatorial model for F^μ at $q = 1$, one has $\pi K = \pi$.*

Remark 2. *The conjecture above is precisely the missing strand needed to close Ledger B: it would provide a Markov chain whose transitions are specified by $g(\cdot)$ and the local coefficients a_μ^ν , without direct reference to the polynomials F^μ , while still proving π is stationary.*

6 Incomplete proof. Logs for reference

6.1 Ledger A (existence + stationarity)

Mechanism M_1 is accepted with:

$$\text{Completeness} = 0.99, \quad \text{Invariance} = 0.99.$$

The proof is fully resolved (Sections 4.1–4.4).

6.2 Ledger B (strict nontriviality)

The best candidate toward strict nontriviality (Section 6) remains incomplete:

$$\text{Completeness} = 0.60, \quad \text{Invariance} = 0.70,$$

reflecting that row-stochasticity and support-marginal matching are proven (under explicit assumptions), but full stationarity of π for that polynomial-free kernel is not proven in this report.